

Filtering noise in time and frequency domain for ultrafast pump-probe performed using low repetition rate lasers

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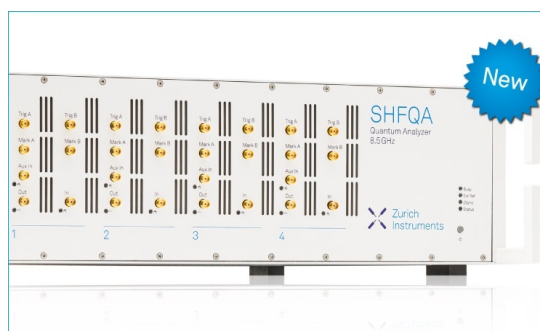
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ABSTRACT

Optical pump-probe spectroscopy is a powerful tool to directly probe the carrier dynamics in materials down to sub-femtosecond resolution. To perform such measurements, while keeping the pump induced perturbation to the sample as small as possible, it is essential to have a detection scheme with a high signal to noise ratio. Achieving such a high signal to noise ratio is easy with phase sensitive detection based on a lock-in-amplifier when a high repetition rate laser is used as the optical pulse source. However, such a lock-in-amplifier based method does not work well when a low repetition rate laser is used for the measurement. In this article, a sensitive detection scheme, which combines the advantages of a boxcar that rejects noise in time domain and a lock-in-amplifier that isolates the signal in the frequency domain for performing pump-probe measurements using a low-repetition rate laser system, is proposed and experimentally demonstrated. A theoretical model to explain the process of signal detection and a method to reduce the pulse to pulse energy fluctuation in probe pulses is presented. By performing pump-probe measurements at various detection conditions, the optimum condition required for obtaining the transient absorption signal with low noise is presented. The reported technique is not limited to pump-probe measurements and can be easily modified to suit for other sensitive measurements at low repetition rates.

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I. INTRODUCTION

Ultrafast optical pump-probe spectroscopy is a powerful tool to directly study the carrier dynamics in materials. In the simplest form of this technique, a high energy pump pulse, which has a temporal width shorter than the response time of the material, is used to change the carrier distribution in the sample. This change will modify the optical response of the material.^{1–4} The change in the optical property of the sample is then measured using another time delayed short and low energy probe pulse. The ultrafast dynamics in the sample can be probed by varying the delay between the pump and probe pulses obtained from an ultrafast laser. Since optical pulses of temporal width as short as few tens of atto-seconds are available, it is possible to study any processes that are longer than the pulse width.

To correctly understand the carrier dynamics, it is essential to keep the change induced by the pump to be as minimum as possible.⁵ Owing to this, the typical change in the transmission or reflectivity of the sample is expected to be very small ($\sim 10^{-3}\%$ to $\sim 10^{-7}\%$). Hence, such measurements need a high signal to noise (S/N) ratio.^{5,6} Different types of noises can affect the measurements and reduce the S/N ratio. Typical sources of optical noise in such measurements are the fluctuations in the average power of the laser used in the measurement and any other background light that is being detected by a photodetector.^{7,8} In addition to these optical noises, the photodetector or the electronics used for the detection can pick up electrical noises from various sources.^{9–11} Since the AC power supply has a frequency of 50 Hz, the electronic noise will be generated at this frequency. Furthermore, ordinary laboratory light sources would produce light at a frequency of 100 Hz (double of 50 Hz). The $1/f$

electrical noise will dominate at low frequencies. High frequency switches and RF sources will produce electrical pickup noise in high frequencies. Furthermore, when a ultrafast laser pulse is incident on the photo-diode, there is a sudden increase in the current flow that can be seen at the output of the photo-diode as an immense peak.¹² Although such noises can cause a random fluctuation in the signal at any given instant of time, a spectral analysis of the noise would show that each source will contribute to noise at different frequencies and its harmonics.

One of the commonly used technique for achieving a high signal to noise ratio in the pump-probe measurement is the phase sensitive detection technique that uses a lock-in-amplifier.^{13,14} In this technique, first a frequency at which the noise is least is chosen, which typically lies in few kHz range. The pump beam should be modulated at that frequency. The lock-in-amplifier then measures the signal at the modulation frequency, thus isolating other noises that are at different frequencies resulting in a good signal to noise ratio.^{14,15} Most of the femtosecond oscillators operate at high repetition rates (for example, ~80 MHz). In such cases, mechanical choppers or electro-optic modulators operated at few kHz or few tens of kHz could be used to provide a good signal to noise ratio in the measurement. Typical femtosecond oscillators provide very low output per pulse energies and the pulse to pulse separation is also very short (12.5 ns for a 80 MHz system). In many samples, the carrier relaxation and heat dissipation takes much longer time than the pulse to pulse separation in oscillators. Femtosecond amplifier systems can provide much higher per pulse energies with long pulse to pulse separation (1 ms for 1 kHz operation). If such amplifier systems are used for pump probe studies, the modulation frequency has to be kept below 1 kHz. For obtaining a good S/N ratio, it is essential to have a good separation between the laser repetition rate and the chopping frequency. Too much reduction of chopping frequency is undesirable because at very low chopping frequencies, $1/f$ noise will start dominating.^{9,10} On the other hand, the high-voltage switching circuits operating at the repetition rate of the laser system also produces electrical pickup noise in the detection system. In addition, the pulse to pulse energy fluctuations in the case of 1 kHz systems are much more when compared to high repetition rate lasers. Thus, in the case of low repetition rate lasers, signals are detected in time domain using boxcar or data acquisition cards with lot of averaging to achieve high S/N ratio.^{16–20} Statistically, as the number of averaging (N) is increased, the standard deviation in the data can be reduced as $\sqrt{N}$ (for $N \gg 1$). However, prolonged acquisition time can cause larger deviations in measurement conditions such as temperature, sample condition, and laser power. Such variations can lead to deviation in the measurement of the true optical response of the sample. Furthermore, in many samples, the decay time also depends on the excited carrier density and, hence, the pump power.^{21–23} In such cases, averaging over large number of data with larger deviation in pump power would result in a wrong estimation for the response time.

In this article, a new technique that combines the advantages of both boxcar and lock-in-amplifier has been proposed and experimentally demonstrated by performing transient absorption measurement on a standard sample using a 1 kHz laser in a two-color pump-probe geometry. We also analyze the problem theoretically to explain the process of noise elimination. In this technique, first a boxcar is used to detect the signal reducing the noise in time

domain. The output of the boxcar is then detected by a lock-in-amplifier to isolate noise in frequency domain, thus improving S/N. The reported technique is general and will be useful for performing sensitive measurements at low-repetition rates.

II. THEORETICAL BACKGROUND

Consider a standard transient pump-probe measurement performed on a sample using time synchronized pump and probe pulses (Fig. 1). In such measurements, the change in the absorption/reflectivity of the sample is first measured at a given delay between the pump and probe pulses, and this measurement is then repeated at several other delays to obtain the complete temporal response.^{8,24} Let us consider a situation in which the laser source used in the measurement is highly stable and has sufficiently high pulse repetition rate. If the response time of the photodetector (PD) being used is slow compared to the pulse to pulse separation, then the output voltage of the PD will be a constant, which is proportional to the average power of the laser beam being detected. Let P_{in} be the average power of the probe beam falling on the front side of the sample (for simplicity, reflection losses are neglected in this analysis). The output voltage of the PD detecting the transmitted power through the sample can be written as

$$V_O(t) = CTP_{in} + Re\left[\sum_{\omega_N} V_N(\omega_N)e^{-i\omega_N t}\right], \quad (1)$$

where C is a constant that depends on the sensitivity of the PD and filters that are being used and T is the transmission of the sample. C will have the unit of A^{-1} . The second term accounts for the systematic and random electrical and optical noises in the measurement expressed in the frequency domain. $V_N(\omega_N)$ is the amplitude of the noise at frequency ω_N . When the sample is excited by the pump pulse, its optical response changes. For sufficiently small excitation conditions, the change in the absorption coefficient will be proportional to the third-order nonlinear absorption coefficient.^{2,22} Consider an instantaneously responding sample having a relaxation time much longer than the pulse width, the absorption coefficient after excitation by the pump pulse can be written as

$$\alpha(E_P, \tau) = \alpha_0 + \beta(\tau)E_P, \quad (2)$$

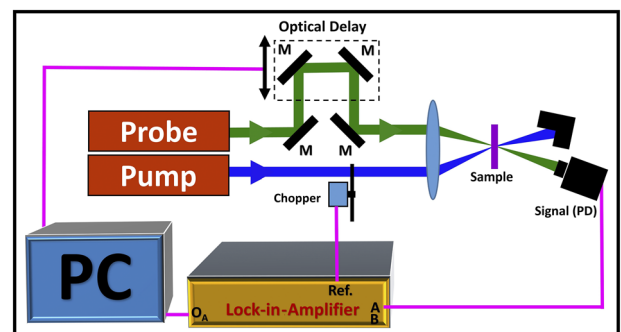


FIG. 1. Schematic of a typical pump-probe setup used for the measurement of transient absorption based phase sensitive detection. M, mirror; PD, photodetector.

where α_0 is the linear absorption coefficient, E_p is the energy of the pump pulse, β is related to the third-order nonlinear absorption coefficient with dimension of α/E_p , and τ is the time between the excitation of the pump pulse and the arrival of the probe pulse. It is the temporal evolution of β that finally reveals the carrier dynamics in the sample. For a sufficiently small induced change in the optical response ($\alpha_0 \gg \beta E_p$), the transmission of the sample in the presence of the pump pulse is given by

$$T = e^{-\alpha d}, \quad (3)$$

$$\approx T_0[1 - \beta(\tau)E_p d], \quad (4)$$

where T_0 is the linear transmission of the sample having thickness d and is equal to $\exp(-\alpha_0 d)$. The change in the transmission of the sample ($\Delta T = T - T_0$) induced by the pump pulse is given by

$$\Delta T(\tau) = -T_0\beta(\tau)E_p d. \quad (5)$$

If the pump beam is modulated at a frequency Ω_c , the transmission of the sample will also change at that frequency between $T_0(1 - \beta E_p d)$ (when the pump pulse is falling on the sample) and T_0 (when the pump is blocked). To get a good S/N ratio, the Ω_c has to be chosen such that the electrical and optical noise in the lab is least at this frequency. Thus, the output voltage of the PD when the pump beam is modulated is given by

$$V_O(t) = CT_0[1 - \beta(\tau)E_p \mathcal{G}(t)d]P_{in} + Re\left[\sum_{\omega_N} V_N(\omega_N)e^{-i\omega_N t}\right]. \quad (6)$$

Here, $\mathcal{G}(t)$ is the pump beam modulating function. Assuming a square wave modulation of the pump pulse, which is typical for the mechanical chopping of the pump beam, the modulating function $\mathcal{G}(t)$ can be expressed in terms of infinite sum of sinusoidal waves given by²⁵

$$\mathcal{G}(t) = \frac{4}{\pi} \sum_{k=1}^{\infty} \frac{\sin[(2k-1)\Omega_c t]}{2k-1}. \quad (7)$$

In a phase sensitive detection technique, the output of the PD is connected to the input of a lock-in-amplifier. Using the chopper frequency as the reference signal, the lock-in-amplifier will measure the amplitude of the voltage at Ω_c . Substituting Eq. (7) into Eq. (6), the output of the lock-in amplifier, V_O , at the frequency Ω_c can be obtained and is given by

$$V_L(\tau) = -\frac{4C}{\pi} T_0\beta(\tau)E_p P_{in} d. \quad (8)$$

As mentioned earlier, the Ω_c is chosen such that there is negligible noise at that frequency, and hence, $V_N(\Omega_c) \approx 0$. Using Eqs. (5) and (8), the transient transmission at the delay τ can be written as

$$\frac{\Delta T(\tau)}{T_0} = \frac{\pi}{4C} \frac{V_L(\tau)}{P_{in} T_0}. \quad (9)$$

A direct measurement of transmitted probe power through the sample using the same PD under similar conditions by blocking the pump beam would produce a constant voltage, $V_0 = CT_0 P_{in}$, at the output of the PD. Using this, Eq. (9) can now be written as

$$\frac{\Delta T(\tau)}{T_0} = \frac{\pi}{4} \frac{V_L(\tau)}{V_0}. \quad (10)$$

Thus, by using the output of the lock-in-amplifier (V_L) at various delays and V_0 , the transient transmission through the sample could be measured with a high S/N ratio.

The repetition rate of several femtosecond oscillators is of the order of 80 MHz, which corresponds to a pulse to pulse separation of 12.5 ns. Typical PDs that have microsecond response times will give a DC output when irradiated with an output of such high repetition rate lasers. By chopping the pump beam at few kHz, which is well away from the laser repetition rate and normal sources of noises, a good signal to noise ratio for the transient measurement could be obtained.^{22,26} As mentioned in the introduction, many samples need higher pump pulse energies for generating sufficient changes in the probe as well as long pulse to pulse separation for providing sufficient time for the sample to relax. Femtosecond amplifier systems can provide much higher energies and are generally operated at ~ 1 kHz, which corresponds to a pulse to pulse separation of 1 ms. When pump-probe measurements are carried out using such beams, the separation between the chopping frequency and laser repetition rate is very less. Furthermore, $1/f$ noise and other unwanted electrical and optical noises become difficult to isolate when the chopper frequency is less than 1 kHz. When low repetition rate lasers are used for a pump-probe measurement, the true signal arrives at the photodetector only for a very short duration, while rest of the time (much longer than the signal duration) the photodetector generates an unwanted signal from the dark current, stray light, and other electrical noises. Such an unwanted accumulation of noise can be removed if the signal is measured only at the time of pulse arrival and the rest of the data are rejected. Boxcar is an electrical instrument that can detect the signal from a photodetector for a specific duration and produce a corresponding DC voltage at its output. The boxcar can be triggered by using a voltage pulse and it is possible to set a delay between the trigger and the measurement of signal. Boxcar also has a provision to average the measured signal. P. Bartolini *et al.* reported a real-time acquisition system in which a balanced detector (for minimizing common-mode noise), a boxcar averager, and a data acquisition board were used for pump-probe measurement using a low-repetition rate laser.¹⁷ For pump-probe measurements with a high-repetition rate laser, they have replaced the boxcar with a lock-in-amplifier. Acquisition of large amount of data using data acquisition cards and post-processing of the collected information with lot of averaging has also been used for improving the signal to noise ratio.¹⁷⁻²⁰ Here, we present a new detection technique by utilizing the merits of both the boxcar and lock-in-amplifier instruments and the same is shown to be very useful in the case of pump-probe experiments performed with a low repetition rate (1 kHz) laser system.

III. EXPERIMENTAL DETAILS

For the demonstration of the present detection technique that combines the advantages of boxcar and lock-in-amplifier, pump-probe measurement was carried out on a silver nanoparticle film by exciting at 400 nm and probing at 408 nm. Figure 2 shows the schematic diagram of the experimental setup along with the details of the detection scheme. The schematic of the pump-probe setup remains the same as that shown in Fig. 1 except for the detection of reference, but with a very different detection scheme, which also incorporates a boxcar. The femtosecond oscillator-amplifier system

where N is the number of data points, S_i represents the individual measured data, and S_{av} is the average defined by

$$S_{av} = \frac{1}{N} \sum_{i=1}^N S_i. \quad (12)$$

Since the data are normalized to the average, the lower value of S_D implies a good S/N ratio. The S_D estimated for the signal measured at the peak of $\Delta T/T$ shown in Fig. 3(a) is 27 ($N = 100$).

If the pulse to pulse power fluctuation in the laser is negligible, the collected data can be considered as similar to that of a standard transient transmission measurement performed using a high-rep rate laser. Generally, the pulse to pulse energy stability of the femtosecond amplifier system is much poorer when compared to that of oscillators. Thus, the transient signal measured using this low repetition rate oscillator–amplifier–OPA system will have additional noise due to the pulse to pulse energy fluctuation. In the following, we show first theoretically and then experimentally the effect of such pulse to pulse energy variation in the probe pulse on the S/N ratio and a method to counter it.

Let the average pulse energy of the probe pulse at the input of the sample be E_{in} with a random pulse to pulse energy fluctuation ΔE_{in} . The value of ΔE_{in} would change at every 1 ms in a 1 kHz repetition rate system. The distribution of ΔE_{in} for an ensemble of pulses would be a Gaussian, which is typical for the type of laser used in the present measurement. The voltage output of the boxcar channel, which is detecting the transmitted probe pulse through the sample using PD₁, is

$$V_O^B(t) = C^B T [E_{in} + \Delta E_{in} \mathcal{H}(t)] + Re \left[\sum_{\omega_N} V_N^B(\omega_N) e^{-i\omega_N t} \right], \quad (13)$$

where C^B is a constant that depends on the sensitivity of the PD₁ and gain in the boxcar. V_N^B is the amplitude of the electrical and optical noises at frequency ω_N . Once again, for a small change in the transmission of the sample in the presence of the pump pulse (E_p) modulated by a function $\mathcal{G}(t)$, the output of the boxcar can be written as

$$V_O^B(t) = C^B T_0 [1 - \beta(\tau) E_p \mathcal{G}(t) d] [E_{in} + \Delta E_{in} \mathcal{H}(t)] + Re \left[\sum_{\omega_N} V_N^B(\omega_N) e^{-i\omega_N t} \right]. \quad (14)$$

Here, $\mathcal{H}(t)$ is a unit step function that repeats with the arrival of the probe pulse. Since ΔE_{in} varies randomly depending on the energy of each probe pulse, the product of ΔE_{in} and $\mathcal{H}(t)$ will be composed of several frequencies and can also have a component at the chopper frequency.

For sufficiently small fluctuations in the probe energy $\Delta E_{in} \ll E_{in}$, the term containing the product of β and ΔE_{in} in Eq. (14) can be neglected when compared to the other terms. Thus, the output of the lock-in-amplifier, which is now detecting the BC output in channel A, can be written as

$$V_L^B(\tau) = -\frac{4C^B}{\pi} E_{in} T_0 \beta(\tau) E_p d + \Delta E_{in} \mathcal{F}(t) C^B T_0, \quad (15)$$

where $\mathcal{F}(t)$ is the amplitude of the Fourier transform of $\Delta E_{in} \mathcal{H}(t)$ at the chopping frequency, and it depends on time because of the slow variations in the laser pulse energy. Note that the second term

in Eq. (15) does not depend on the $\beta(\tau)$ of the sample but depends directly on the pulse to pulse fluctuation. Thus, due to the shot-to-shot energy fluctuation in the probe, additional noise [given by the second term in Eq. (15)] will get added to the required signal, the first term. The $\Delta T/T$ presented in Fig. 3 will have fluctuations caused by the random probe energy variation.

For removing this noise, a small part of the probe beam was reflected before the sample and was detected using another photodetector (PD₂). The output of the PD₂ is also processed by another channel of the boxcar. The output of the boxcar can be written as

$$V_O^{B2}(t) = C^{B2} T_f R [E_{in} + \Delta E_{in} \mathcal{H}(t)] + Re \left[\sum_{\omega_N} V_N^{B2}(\omega_N) e^{-i\omega_N t} \right], \quad (16)$$

where C^{B2} is a constant that depends on the sensitivity of the PD₂ and gain in boxcar. R is the reflectance of the beam splitter used for obtaining the reference beam and T_f is the transmission of the filter used before PD₂. Using a variable filter, the T_f was chosen such that $C^{B2} R T_f = C^B T_0$ (one can also change the gain in the boxcar channel detecting the output of PD₂). This can be easily done directly by monitoring the boxcar outputs $V_O^B(t)$ and $V_O^{B2}(t)$ in an oscilloscope and making them almost equal. This voltage $V_O^{B2}(t)$ is then subtracted to the boxcar output of PD₁ [$V_O^B(t)$] electronically before feeding in to the lock-in-amplifier. Most of the commercial lock-in-amplifier do come with a provision for such subtraction (channel B) of voltage, which could be directly used for this subtraction purpose. The output voltage after subtraction (A–B) can be written as

$$V_O^D(t) = -C^B T_0 \beta(\tau) E_p \mathcal{G}(t) d [E_{in} + \Delta E_{in} \mathcal{H}(t)] + Re \left[\sum_{\omega_N} (V_N^B - V_N^{B2}) e^{-i\omega_N t} \right]. \quad (17)$$

The output of the lock-in-amplifier that is now detecting the $V_O^D(t)$ using the chopping frequency as the reference will be

$$V_L^B(\tau) = -\frac{4C^B}{\pi} E_{in} T_0 \beta(\tau) E_p d. \quad (18)$$

A direct measurement of the transmitted probe pulse energy through the sample using the same PD₁ under similar conditions through the boxcar by blocking the pump beam would produce a constant voltage, $V_D^B = C^B T_0 E_{in}$ at the output of the boxcar. Now using Eq. (5) and the measured V_D^B , the transient transmission through the sample can be estimated by

$$\frac{\Delta T(\tau)}{T} = \frac{\pi}{4} \frac{V_L^B(\tau)}{V_D^B}. \quad (19)$$

It is important to note that in the present technique, the usage of an identical balanced detector is not essential as done by several earlier reports. Normal photodetectors can be used for measuring the reference and signal to reduce the noise due to probe energy fluctuations. Only the DC output of boxcar needs to be made equal [$V_O^B(t) = V_O^{B2}(t)$].

Figure 3(b) shows the measured transient transmission signal of the metal thin film sample performed by the A–B configuration. A single exponential fit to the decay part of this signal gives a decay time of 3.5 ± 0.1 ps. The S_D of the signal measured at the peak change in the present case is 10, which is nearly one-third of that when the

signal measured without the subtraction of boxcar output of PD₂ from that of PD₁.

To further reduce the statistical fluctuations in the measurement, a chopper frequency dependent study was carried out on the same sample. The transient signal at the delay where the peak occurs is recorded repeatedly in the A-B configuration (N is ~ 100) at various chopper frequencies. The statistical variation in the peak signal as given by Eq. (11) was estimated at each chopper frequencies and is given in Table I. Furthermore, apart from S_D , it is also important to know about the maximum and minimum signals obtained in the measured data to understand the stability of the measured signal. Figure 4 shows the variation of S_D on the chopping frequencies as a solid rectangle. Figure 4 also shows the measured maximum and minimum signal values among the dataset at each chopper frequency as bars. Clearly, the S_D is least for the chopping frequency 333 Hz. As the chopper frequency deviates away from 333 Hz, we find that the S_D and the peak to peak variation in the signal also increases.

When the laser pulse repetition rate is very large compared to the chopping frequency, the number of laser pulses blocked or unblocked in each cycle would remain nearly the same. On the other

TABLE I. The dependence of S_D and peak to peak variation in the signal at different chopper frequencies Ω_c around 333 Hz.

Ω_c	S_D (%)	Peak to peak fluctuation (%)
230	15.5	95.3
280	13.9	56.6
333	2.9	14.5
380	11.5	44.6
420	11.6	61.2

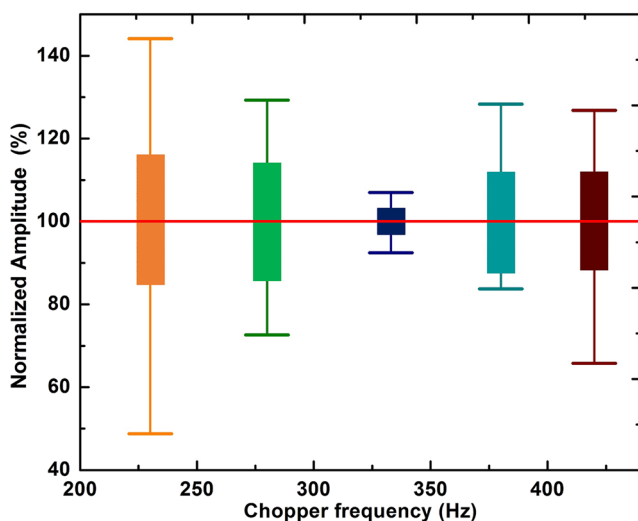


FIG. 4. The dependence of the fluctuations in the normalized signal measured at a peak position in the A-B configuration on the chopping frequencies around 333 Hz. The height of the rectangular box represents the S_D , and the maximum and minimum signals values in the data set are also shown as bars.

hand, when the laser repetition rate and the chopper frequency are close, a temporal aliasing process can lead to frequent changes in the number of pump pulses that are actually getting passed through the chopper. To measure the effect of the chopper on the pump pulses, we have directly measured the energy of the pump pulses after passing through the chopper. Figure 5 shows the output of the boxcar measuring the output of the photodetector detecting the modulated pump-pulses passing through the chopper for 30 cycles. Here, zero amplitude means that the chopper had blocked the pulse while the higher amplitudes correspond to the energy of the pulses passing through the chopper. Clearly, the pump pulses are getting partially blocked by the chopper at frequencies other than 333 Hz. This phenomenon can also be explained as follows. The chopper frequency 333 Hz is a factor of the laser repetition rate, 1 kHz. Thus, in each chopper cycle, two pulses are passed and one pulse is blocked and the process repeats (the reverse, two pulses are blocked and one pulse is passed is also possible depending on the starting time of chopper). On the other hand, if the chopper frequency is such that it is not a factor of the laser repetition rate, then frequently the chopper blade would be in a partially closed or partially open condition during the pulse arrival. Such partial cutting of the pump pulse would lead to strong changes in the pump energy as well as spatial shape distortion of the beam profile at the sample place. Since the transient signal is proportional to the pump pulse energy [see Eq. (18)], these variations would cause the signal to fluctuate. Thus, 333 Hz is one of the possible factor of 1 kHz, which can have the least fluctuation in the signal as well as the least variation in peak to peak. Hence, to avoid such chopping generated modulations in the pump pulse, it is essential to choose the chopping frequency as one of the factors of the laser repetition rate.

We have also performed similar measurements at all the chopper frequencies that are factors of 1 kHz in the A-B configuration. The statistical variations in the peak signal are given in Table II. Figure 6 shows the dependence of S_D and the measured maximum and minimum signal values on the harmonic chopping

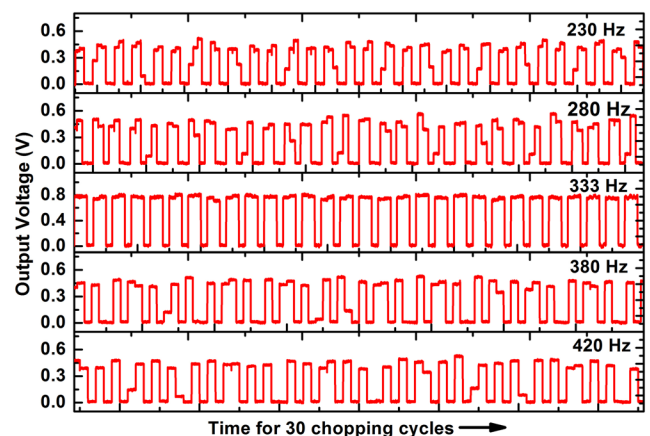
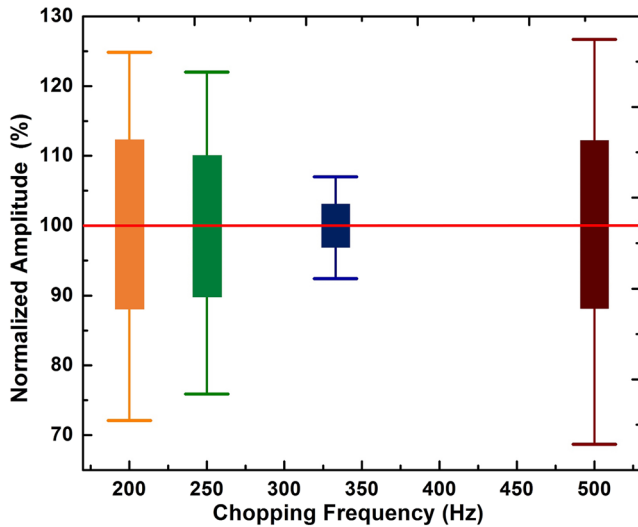


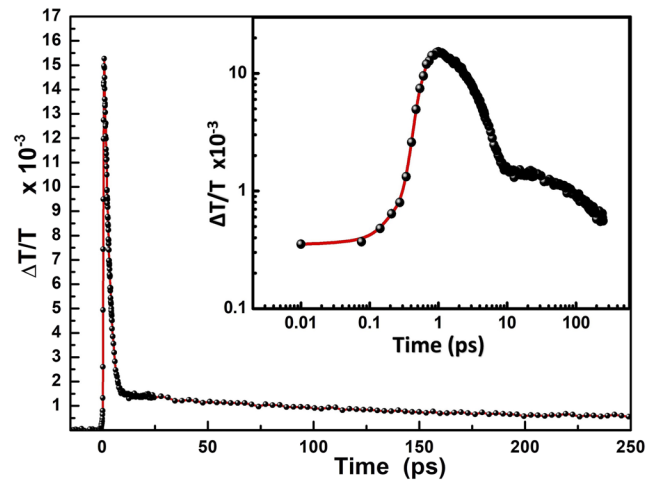
FIG. 5. Output voltage of the boxcar, which is measuring the output of the photodetector sensing pump pulses passing through the chopper operated at different frequencies. The time in the horizontal axis at each chopping frequency is chosen such that there are 30 chopping cycles in view.

TABLE II. The dependence of S_D and peak to peak variation in the signal at different chopper frequencies, which are factors of 1 kHz.

Ω_c	S_D (%)	Peak to peak fluctuation (%)
200	12.0	52.8
250	9.8	46.2
333	2.9	14.5
500	12.0	58.0

**FIG. 6.** The dependence of the fluctuations in the normalized signal measured at the peak position in the A-B configuration on the chopping frequencies, which are factors of 1 kHz. The height of the rectangular box represents the S_D and the maximum and minimum signal values in the dataset are also shown as bars.

frequencies, which are factors of 1 kHz. Once again, the S_D for the case of chopping frequency 333 Hz has the least S_D . Although the chopper frequencies 200 Hz, 250 Hz, and 500 Hz are factors of 1 kHz, still the noise at these frequencies are much higher than that of 333 Hz. Pump-probe measurements performed at very low chopper frequencies like 50 Hz or 100 Hz (factors of 1 kHz) result in increasing the shot-to-shot fluctuations leading to even lower signal-to-noise ratios. The AC power supply frequency used in the experiment is 50 Hz, and the on-off frequency of most of the laboratory light sources driven by this AC power supply is 100 Hz. Thus, because of any kind of stray light present in the lab, it is well known that 50 Hz, 100 Hz, and their higher harmonics are all not suitable for chopping frequencies. When chopped at these frequencies, the lock-in-amplifier will sense these electronic noises or the other light that is being detected by the photodetectors. Since the chopping frequencies 200 Hz, 250 Hz, and 500 Hz are all higher harmonics of 50 Hz and 100 Hz, the signal detected at these frequencies will have higher noise compared to that at 333 Hz. Several earlier reports used 500 Hz or various other chopping frequencies for the measurement of the pump-probe signal with a 1 kHz system.^{19,20,29} Our results suggest that for a best S/N ratio, the pump should be chopped at 333 Hz.

**FIG. 7.** A typical transient absorption signal measured with 333 Hz chopping. Inset shows the same curve in the log-log scale.

Furthermore, even for the case where only a lock-in-amplifier is used in the measurement, the chopping has to be done at 333 Hz. Although the technique reported here takes care of the fluctuations in data caused by the probe energy variation, still the variations in pump energy would cause the signal to vary [Eq. (18)]. Figure 7 shows a transient absorption signal measured on a silver nanoparticle sample under fully optimized conditions and averaged over eight measurements at each delay. In such a long delay range, a second slow decay in the transient signal, which is due to the phonon-phonon interaction, can also be clearly identified (see the inset of Fig. 7).^{22,27}

IV. CONCLUSION

In this work, a detection scheme that combines the advantages of boxcar and lock-in-amplifier for performing the pump-probe using the low-repetition rate laser system was demonstrated. A theoretical model was developed to explain the process of signal detection. The boxcar was operated such that it detects the peak signal from the photodetectors, avoiding any noises that are being picked up by the photodetectors, thus isolating noise in the time domain. Chopping the pump and detecting the output of the boxcar using a lock-in-amplifier provides a way to isolate other electrical and optical noises that generally appear at other frequencies. In addition, these noise isolations, a way to reduce the signal fluctuations due to pulse to pulse probe energy variation, were also shown experimentally. A chopping frequency dependent study shows that for the best signal to noise ratio, the chopper frequency should be a factor of the laser repetition rate. Furthermore, it should not also be a harmonic of the AC power supply. Thus, in the case of low-repetition rate lasers, it is essential to move away from not only harmonics of AC supply but also to a frequency, which is a factor of its repetition rate. Furthermore, we also find that the gain in the boxcar could also be used to increase the signal without much change in the S/N ratio.

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DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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